

EE 474 Introduction to Control Systems**Homework 4**

Due Wednesday, October 7, 2026

Reading:

- Franklin *et al.*, Chapter 3, Introduction and Section 3.1

1 Text problemsA) Text problem 2.20 (2.22 in 8th Ed.)

Text problem 2.20 :

A very typical problem of electromechanical position control is an electric motor driving a load that has one dominant vibration mode. The problem arises in computer-disk-head control, reel-to-reel tape drives, and many other applications.

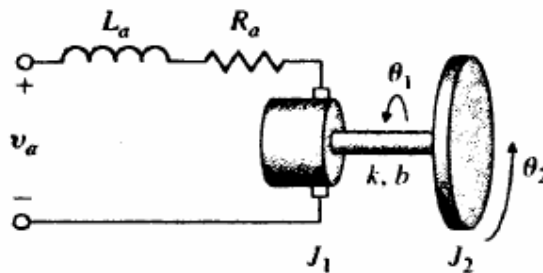
- A schematic diagram is sketched in Fig. 2.54. The motor has an electrical constant K_e , a torque constant K_t , an armature inductance L_a and a resistance R_a

The rotor has an inertia J_1 and a viscous friction B . The load has an inertia J_2 . Note: B is not shown in the diagram. B creates a torque on J_1 according to

$$T_B(t) = B \dot{\theta}_1(t)$$

The two inertias are connected by a shaft with a spring constant k and an equivalent viscous damping b . Write the equations of motion.

Figure 2.54
Motor with a flexible load



A.1) Write the equations of motion (as asked for in the problem). The transfer function is not required at this stage. Note that there will be 3 differential equations:

- One for J_1 velocity $\dot{\theta}_1$
- One for J_2 velocity $\dot{\theta}_2$
- One for current I_a

A.2) Form the analogous circuit model for the mechanical sub-system of text problem 2.20 (the two inertias, with motor torque as the input).

- Hint: the circuit will have two nodes, corresponding to the two velocities, and a current source, corresponding to the motor torque.
- Label each circuit element, and give the value of each circuit element in terms of the component values in the original system.
 - Examples: $C_1 = J_1$, $R_1 = 1/B$

B) Show that units balance in equation (2.62)

$$J_m \ddot{\theta}_m + b \dot{\theta}_m = K_t i_a \quad (\text{Text, eqn (2.62)})$$

C) Nothing real is perfectly linear, but of all types of systems:

- Electrical circuits are most linear, a resistor can linear to a part in 10^8
- Fluid systems are often least linear (such as a hydraulic drive for a plastic injection molding machine).

Components such as valves are quite nonlinear, and controls are designed about an operating point (e.g., a power plant might have controls designed for operating at 50% of capacity, 80% of capacity and 100%).

The following is a model for fluid flow through a valve

$$q(t) = 20 \frac{e^{x(t)/15} - 1}{e^{x(t)/15} + 1} \quad (1)$$

where $q(t)$ is the flow in m^3/second , and $x(t)$ is the valve angle in degrees.

C.1) Using Matlab, plot the input-output relationship for x over the range

$$0 \leq x \leq 90$$

Add a grid, labels and title to your plot.

C.2) Using the technique presented in Lab Part 1, "Linearization of a nonlinear plant" determine the linearized model for flow versus valve angle when the steady state valve angle is

C.2.a) 20 deg.

C.2.b) 60 deg.